

The growth-law correction constant $\delta = \log R_\infty$ of the quadratic polynomial continued fraction: structure, a closed-form bracket, and certified high-precision values

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June 2026 (`pcf-delta` v1.4 draft)

Abstract

For integers $A, C \geq 1$ and $B \geq 0$ let $b_k = Ak^2 + Bk + C$ and let q_n be the standard continuant denominators of the polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n \geq 1} 1/(An^2 + Bn + C)$ ($q_0 = 1$, $q_1 = b_1$, $q_n = b_n q_{n-1} + q_{n-2}$). A companion note proves the growth law $\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + (K_\Gamma + \delta) + \kappa/n + O(1/n^2)$, in which the product constant $K_\Gamma = -\log(\Gamma(1-r_1)\Gamma(1-r_2))$ is closed-form but the correction $\delta = \log R_\infty > 0$, with $R_\infty = \lim_n q_n / \prod_{k \leq n} b_k$, is left undetermined.

This note *characterizes* δ . We show that the scaled limit R_∞ is exactly the *independence polynomial of a path graph* evaluated at the weights $u_n = 1/(b_{n-1}b_n)$, giving a cluster expansion $R_\infty = \sum_{k \geq 0} \sigma_k$ with $\sigma_k \leq S^k$, $S = \sum_{n \geq 2} u_n$. From it we derive a closed-form two-sided bracket $\log(1+S) \leq \delta \leq -\log(1-S)$, with S in closed digamma form (generic poles) and confluent polygamma form (degenerate poles; e.g. $S(1, 2, 1) = \pi^2/3 - 13/4$ exactly). Using the first three cluster sums $\sigma_1, \sigma_2, \sigma_3$ in exact closed form (machine-verified against brute-force enumeration to $\sim 10^{-54}$) to seed an exact downward recurrence, we evaluate δ to certified high precision: $\delta(1, 0, 1)$ to ≥ 44 digits, with a nine-triple table, each value certified two internal ways and cross-checked by a fully independent forward/Neville extrapolation. An integer-relation (PSLQ/`identify`) search returns no low-height closed form, and a precision-stability test shows the uncontrolled relations to be numerical artifacts; we record δ as conjecturally a new transcendental constant of the family. Finally, the finitary structural core — the Casoratian identity, the monotone-bounded convergence $R_n \rightarrow R_\infty$, the closed-form bracket $1+S \leq R_\infty \leq 1/(1-S)$, and the independence-polynomial identity $R_n = \sum_T \prod_{i \in T} u_i$ — is machine-checked in Lean 4/Mathlib with clean axiom cones (no `sorry`); Versions 1.3–1.4 enlarge this core (Section 8) with a general-ring Casoratian, a degree-graded refinement carrying the σ_3 closed form, an m -term geometric-tail enclosure, and exact weighted telescoping for the $B = 0$ family.

2020 MSC: 11J70 (primary), 11A55, 33B15, 05C31. **Keywords:** polynomial continued fraction, convergent denominators, independence polynomial, cluster expansion, digamma function, high-precision computation, integer-relation detection.

1 Introduction

The convergent denominators of the quadratic polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n \geq 1} 1/(An^2 + Bn + C)$, integers $A, C \geq 1$, $B \geq 0$, obey $q_0 = 1$, $q_1 = b_1$, $q_n = b_n q_{n-1} + q_{n-2}$ with

$b_k = Ak^2 + Bk + C$. A companion note [1] establishes the exact two-sided growth law

$$\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + (K_\Gamma + \delta) + \frac{\kappa}{n} + O(1/n^2), \quad (1)$$

where r_1, r_2 are the roots of $Ax^2 + Bx + C$, $K_\Gamma = -\log(\Gamma(1-r_1)\Gamma(1-r_2))$ is the *product constant* (at $B = 0$ equal to $\log(\sinh(\pi\sqrt{C/A})/(\pi\sqrt{C/A}))$), $\kappa = \frac{1}{2}(B^2/A^2 + B/A - 2C/A)$ (which arises *solely* from the Euler–Maclaurin tail of the product $\log P_n = \sum_{k \leq n} \log b_k$, *not* from δ), and

$$\delta = \log R_\infty > 0, \quad R_\infty = \lim_{n \rightarrow \infty} \frac{q_n}{\prod_{k=1}^n b_k} \quad (2)$$

is the *correction constant* produced by the $+q_{n-2}$ term of the recurrence. The product piece K_Γ is closed-form; the companion note claims *no* closed form for δ and treats it as a black box. This note opens the box. For self-containment we stress that the growth law (1) enters only as motivation: every result below concerns $R_\infty = \lim_n q_n / \prod_{k \leq n} b_k$ directly and is logically independent of (1), which is itself openly deposited [1].

Our contributions are, in increasing order of analytic depth:

- (i) **Structure.** R_∞ is exactly the independence polynomial of the path graph on $\{2, \dots, n\}$ in the limit, evaluated at $u_n = 1/(b_{n-1}b_n)$ (Proposition 1); equivalently a Mayer-type cluster expansion $R_\infty = \sum_k \sigma_k$ with $\sigma_k \leq S^k$.
- (ii) **A closed-form bracket.** $\log(1+S) \leq \delta \leq -\log(1-S)$ (Proposition 2), where $S = \sum_{n \geq 2} u_n$ is closed-form via a digamma residue sum, with a confluent polygamma form at degenerate poles; one degenerate value is exactly rational-plus- π^2 : $S(1, 2, 1) = \pi^2/3 - 13/4$.
- (iii) **Certified high precision.** Exact closed forms for $\sigma_1, \sigma_2, \sigma_3$ seed an exact downward recurrence, yielding δ to ≥ 44 digits for $(1, 0, 1)$ and a nine-triple table (Section 5); each value is certified by seed-order independence and self-convergence, and cross-checked by an independent extrapolation method.
- (iv) **Non-elementarity (conjectural).** An integer-relation search finds no low-height closed form, and a precision-stability test exhibits the uncontrolled relations as artifacts (Section 6).
- (v) **A machine-checked core.** The finitary structural facts on which (i)–(iii) rest — the Casoratian identity, the monotone-bounded convergence $R_n \rightarrow R_\infty$, and the closed-form bracket $1+S \leq R_\infty \leq 1/(1-S)$ (conditional on $\sum u_n < \infty$ and $S < 1$) — are formalized in Lean 4/Mathlib with clean axiom cones (Section 7). Section 8 adds v1.3–v1.4 generalizations and refinements (a general Casoratian over any commutative ring, a degree-graded independence refinement and the σ_3 closed form, an m -term enclosure, and exact weighted telescoping for $B = 0$), keeping every analytic input an explicit hypothesis.

Relation to prior work. The denominators q_n satisfy the three-term recurrence $q_n = b_n q_{n-1} + q_{n-2}$ with polynomially growing coefficients $b_n = An^2 + Bn + C \rightarrow \infty$, a Poincaré-type difference equation. Such a recurrence has a two-dimensional solution space spanned by a *dominant* solution (here q_n) and a *recessive* (minimal) solution; convergence of the continued fraction to V is exactly the selection of the minimal ray by p_n/q_n , the content of Pincherle’s theorem (originally due to Pincherle; modern statement and attribution in [3], with minimal/recessive solutions in [7, §3.6]). The leading large- n asymptotics of growing-coefficient recurrences are governed by Perron’s theorem [4] and the Birkhoff–Trjitzinsky/Wong–Li asymptotic theory [5, 6]; applied to q_n these produce precisely the

dominant terms $n \log A + 2 \log(n!) + \frac{B}{A} \log n$ and the Gamma-product constant K_Γ of the companion law (1). What that theory leaves undetermined is the additive *constant* — the amplitude relating the recessive solution to the product scale $\prod_k b_k$. The present note isolates and characterizes that constant, $\delta = \log R_\infty$: we show R_∞ is the path independence polynomial at the weights $u_n = 1/(b_{n-1}b_n)$ (Proposition 1), from which the closed-form bracket and the certified values below follow. To our knowledge neither this combinatorial identity for the amplitude nor the digamma closed form for its bracketing series S appears in that literature.

Standing assumptions. Throughout A, C are integers ≥ 1 and B is an integer ≥ 0 , so $b_k = Ak^2 + Bk + C \geq Ak^2 + C \geq A + C \geq 2 > 0$ for all $k \geq 1$, $q_n > 0$, and $P_n := \prod_{k=1}^n b_k > 0$. The roots of $Ax^2 + Bx + C$ satisfy $r_1 + r_2 = -B/A \leq 0$ and $r_1 r_2 = C/A > 0$, hence form either a complex-conjugate pair (purely imaginary exactly when $B = 0$) or two negative reals; in all cases $\operatorname{Re} r_i \leq 0$.

Epistemic convention. We use the four-class SIARC partition. **PROVEN** = machine-checked in Lean with a clean axiom cone (a subset of `{propext, Classical.choice, Quot.sound}` and no `sorryAx`); **STRUCTURAL** = complete elementary hand proof, not yet formalized; **VERIFIED** = confirmed numerically for the stated finite cases; **CONJECTURED** = supported by evidence, not established. Each numbered result below is tagged accordingly, and Section 9 collects the grading. All numerical claims are reproduced by the scripts named in Section 10.

2 The scaled recurrence and the Casoratian frame

Let $R_n := q_n/P_n$, so $R_0 = R_1 = 1$. The convergent numerators p_n ($p_{-1} = 1, p_0 = 1$) satisfy the same second-order recurrence; set $P_n^{\text{num}} := p_n/P_n$.

Lemma 1 (Scaled recurrence; STRUCTURAL (tail rate); strict-increase sub-fact PROVEN (`rseq.strictMono.succ`, Section 8), convergence core in Section 7). *For all $n \geq 2$,*

$$R_n = R_{n-1} + u_n R_{n-2}, \quad u_n := \frac{1}{b_{n-1}b_n} > 0.$$

Consequently (R_n) is strictly increasing, bounded above by $\prod_{k \geq 2} (1 + u_k) < \infty$, and converges to a limit $R_\infty \in (1, \infty)$ with $R_\infty - R_n = O(1/n^3)$; thus $\delta = \log R_\infty > 0$.

Proof. Divide $q_n = b_n q_{n-1} + q_{n-2}$ by $P_n = b_n P_{n-1} = b_n b_{n-1} P_{n-2}$ to get $R_n = R_{n-1} + R_{n-2}/(b_{n-1}b_n)$. Since $u_n > 0$ and $R_0, R_1 > 0$, induction gives $R_n > 0$ and the increment $R_n - R_{n-1} = u_n R_{n-2} > 0$, so (R_n) strictly increases; monotonicity gives $R_{n-2} \leq R_{n-1}$, whence $R_n \leq R_{n-1}(1 + u_n)$ and $R_n \leq \prod_{k=2}^n (1 + u_k)$. As $b_{k-1}b_k \geq A^2(k-1)^2 k^2$, the series $\sum_k u_k = O(\sum k^{-4})$ converges, so the product converges and (R_n) is bounded; a monotone bounded sequence converges to $R_\infty \geq R_2 = 1 + u_2 > 1$. The tail estimate $R_\infty - R_n = \sum_{m > n} (R_m - R_{m-1}) = \sum_{m > n} u_m R_{m-2} \leq R_\infty \sum_{m > n} u_m = O(1/n^3)$, where the first equality telescopes the increment $R_m - R_{m-1} = u_m R_{m-2}$, follows by comparison with $\int_n^\infty t^{-4} dt$ since $u_m \sim (Am^2)^{-2}$. $\square$

Lemma 2 (Casoratian; PROVEN, Section 7). *The discrete Wronskian of p_n, q_n is $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$; hence, in scaled variables,*

$$P_n^{\text{num}} R_{n-1} - P_{n-1}^{\text{num}} R_n = \frac{(-1)^{n-1}}{b_n \left(\prod_{k=1}^{n-1} b_k \right)^2}.$$

The initial data $p_{-1}=1, p_0=1, q_{-1}=0, q_0=1$ give $p_1=b_1+1$ and $q_1=b_1$, so Lemma 2 is exactly the convergent Casoratian $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$ of Theorem 1 re-indexed by $n \mapsto n+1$; the Lean statement uses that shifted index and the equivalent sign $(-1)^n$.

The Wronskian decays like $P_n^{-2} \asymp (n!)^{-4}$, which is why the *value* $V = \lim p_n/q_n$ converges super-geometrically (matched to 77–117 digits across the test triples already at $N \approx 4000$, and to the 129 digits used in Section 6 at larger N) while $R_\infty - R_n$ decays only like $1/n^3$. Consistently, the $\kappa/n + O(1/n^2)$ remainder in (1) is the Euler–Maclaurin tail of $\log P_n = \sum_{k \leq n} \log b_k$, whereas $\log(q_n/P_n) = \delta + O(1/n^3)$ contributes only to the constant $K_\Gamma + \delta$ and a strictly smaller remainder absorbed into $O(1/n^2)$. The slow convergence of R_n is what forces the dedicated high-precision method of Section 5; a naive forward recurrence stalls at ~ 25 –38 digits.

3 R_∞ as an independence polynomial

The recurrence $R_n = R_{n-1} + u_n R_{n-2}$ is *not* the classical continuant recurrence $K_n = x_n K_{n-1} + K_{n-2}$: the weight u_n multiplies the *second-order* term. It is instead the transfer recurrence of an independence polynomial.

Proposition 1 (Independence-polynomial identity; PROVEN (Lean, Theorem 1)). *For every n ,*

$$R_n = \sum_{\substack{T \subseteq \{2, \dots, n\} \\ \text{no two consecutive}}} \prod_{i \in T} u_i,$$

the independence polynomial of the path graph on the vertices $\{2, \dots, n\}$ (edges between consecutive integers), evaluated at vertex weights u_i . Letting $n \rightarrow \infty$,

$$R_\infty = \sum_{k \geq 0} \sigma_k, \quad \sigma_0 = 1, \quad \sigma_1 = S = \sum_{n \geq 2} u_n, \quad \sigma_k = \sum_{\substack{T \subseteq \{2, 3, \dots\} \\ |T|=k \\ \text{gaps} \geq 2}} \prod_{i \in T} u_i. \quad (3)$$

Because each non-consecutive k -subset is one of the unrestricted k -tuples (which double-count order, repeats, and adjacency), $0 \leq \sigma_k \leq S^k$, so the series converges geometrically.

Proof. Both sides satisfy the same recurrence and initial data. The empty set gives the constant term 1 ($R_1 = 1$); a non-consecutive subset $T \subseteq \{2, \dots, n\}$ either omits n (contributing to R_{n-1}) or contains n , in which case it omits $n-1$ and the rest is a non-consecutive subset of $\{2, \dots, n-2\}$ with the factor u_n (contributing $u_n R_{n-2}$). This is exactly $R_n = R_{n-1} + u_n R_{n-2}$, and the $n = 1, 2$ base cases match. The bound $\sigma_k \leq S^k$ is the stated over-count. $\square$

The cluster identity (3) is verified *exactly* (subset dynamic-programming equals R_n for all tested triples and all n) by the script `continuant_verify.py`.

4 A closed-form two-sided bracket

Proposition 2 (Bracket; PROVEN (Lean, conditional core; Section 7), VERIFIED all triples). *With $S = \sum_{n \geq 2} u_n < 1$,*

$$1 + S \leq R_\infty \leq \frac{1}{1 - S} \quad \Longleftrightarrow \quad \log(1 + S) \leq \delta \leq -\log(1 - S).$$

Proof. From $R_\infty = 1 + \sum_{n \geq 2} u_n R_{n-2}$ and $R_{n-2} \geq 1$ comes the lower bound $R_\infty \geq 1 + S$. For the upper bound, by (3) and $\sigma_k \leq S^k$, $R_\infty = \sum_k \sigma_k \leq \sum_k S^k = 1/(1 - S)$ (using $S < 1$: each $b_k = Ak^2 + Bk + C$ is increasing in each of A, B, C for $k \geq 1$, so $u_n = 1/(b_{n-1}b_n)$ and hence $S = \sum_{n \geq 2} u_n$ are maximized at the minimal admissible triple $(1, 0, 1)$, giving $S \leq S(1, 0, 1) \approx 0.1307 < 1$).¹ Apply $\log$. $\square$

The bracket is deliberately coarse: the lower bound keeps only $\sigma_0 + \sigma_1$ and the upper bound replaces each σ_k by its geometric majorant S^k . The exact cluster forms σ_2, σ_3 of Section 5 (Appendix A) sharpen it well past the bracket.

Remark 1 (Machine-checked, cluster-free route). *The R_∞ -form of the bracket is formalized in Lean as `rinf_bracket` (Section 7), conditional only on the elementary analytic input that the partial sums $S_n = \sum_{k=2}^n u_k$ are bounded above and $S = \sup_n S_n < 1$. The formalization does not use the cluster expansion (3): it rests on the two finitary inequalities $1 + S_n \leq R_n$ (`rseq_ge_one_add_psum`) and $(\prod_{k=2}^n (1 + u_k))(1 - S_n) \leq 1$ (`rmaj_mul_one_sub_psum_le_one`), the latter from the exact identity $(1 - S_n) - (1 + u)(1 - S_n - u) = u S_n + u^2 \geq 0$. The δ -form follows by monotonicity of $\log$.*

It remains to put S in closed form. Write $b_n = A(n - r_1)(n - r_2)$, so $1/(b_{n-1}b_n)$ has its poles in the multiset $\{r_1, r_2, 1 + r_1, 1 + r_2\}$.

Proposition 3 (Closed form for S ; VERIFIED to 35–62 digits). *If the four poles are distinct,*

$$S = - \sum_{\rho} \text{Res}_{\rho} \psi(2 - \rho), \quad \text{Res}_{\rho} = \frac{1}{A^2 \prod_{\sigma \neq \rho} (\rho - \sigma)}, \quad (4)$$

with ψ the digamma function (the sum of residues vanishes, $\sum_{\rho} \text{Res}_{\rho} = 0$, because $1/(b_{n-1}b_n)$ is a proper rational function whose denominator degree exceeds the numerator degree by $4 \geq 2$ — equivalently its residue at ∞ is 0 — so the individually divergent ψ -pieces combine convergently). For repeated poles, partial fractions over the multiset of multiplicities m_{ρ} give the confluent form

$$S = - \sum_{\rho} a_{\rho,1} \psi(2 - \rho) + \sum_{\rho} \sum_{j \geq 2} a_{\rho,j} \frac{(-1)^j \psi^{(j-1)}(2 - \rho)}{(j-1)!}, \quad (5)$$

$a_{\rho,j}$ the Laurent coefficients of $1/(b_{n-1}b_n)$ at ρ , with $\sum_{\rho} a_{\rho,1} = 0$.

Equations (4)–(5) are the termwise sums $\sum_{n \geq 2} (n - \rho)^{-1}$ (regularized, $j = 1$) and $\sum_{n \geq 2} (n - \rho)^{-j} = (-1)^j \psi^{(j-1)}(2 - \rho)/(j-1)!$ ($j \geq 2$), standard polygamma series [7, §5.15]. Both are confirmed against direct summation (`mpmath.nsum`) for distinct- and repeated-pole triples (`S_closed_form.py`, `S_confluent.py`). A clean degenerate value: for $(A, B, C) = (1, 2, 1)$, $b_n = (n + 1)^2$ has poles $\{-1, -1, 0, 0\}$ and

$$S(1, 2, 1) = \frac{\pi^2}{3} - \frac{13}{4} = 0.0398681336964528729 \dots, \quad (6)$$

matched to 10^{-40} .² (Under the standing assumptions the poles of $1/(b_{n-1}b_n)$, namely $r_1, r_2, 1 + r_1, 1 + r_2$, each have real part $\leq 1 < 2$ because $\text{Re } r_i \leq 0$; hence no summation index $n \geq 2$ is a pole and the residue and confluent forms apply throughout.) Table 1 shows the bracket on representative triples.

¹Monotonicity, stated cleanly: for fixed $k \geq 1$ the partials $\partial b_k / \partial(A, B, C) = (k^2, k, 1)$ are positive, so $b_k = Ak^2 + Bk + C$ is strictly increasing in each of A, B, C ; hence $u_n = 1/(b_{n-1}b_n)$ and $S = \sum_{n \geq 2} u_n$ are strictly decreasing in each. The admissible family has componentwise-minimal element $(1, 0, 1)$, whence $\sup S = S(1, 0, 1) \approx 0.1307 < 1$.

²Elementarily, here $u_n = 1/(b_{n-1}b_n) = 1/(n^2(n+1)^2) = (\frac{1}{n} - \frac{1}{n+1})^2 = \frac{1}{n^2} - \frac{2}{n(n+1)} + \frac{1}{(n+1)^2}$, so $\sum_{n \geq 1} u_n = \zeta(2) + (\zeta(2) - 1) - 2 = 2\zeta(2) - 3 = \frac{\pi^2}{3} - 3$ (the middle sum telescopes, $\sum_{n \geq 1} 1/(n(n+1)) = 1$); subtracting the $n = 1$ term $u_1 = \frac{1}{4}$ leaves the $n \geq 2$ sum $S(1, 2, 1) = \sum_{n \geq 2} u_n = \frac{\pi^2}{3} - \frac{13}{4}$.

Table 1: The closed-form bracket $\log(1 + S) \leq \delta \leq -\log(1 - S)$ on representative triples; δ from the high-precision method of Section 5.

(A, B, C)	S	$\log(1 + S)$	δ	$-\log(1 - S)$
(1, 0, 1)	0.13066961899	0.12281004	0.12385719436	0.14003204
(1, 0, 2)	0.080228466794	0.077172562	0.077741108481	0.083629973
(2, 0, 1)	0.045720089039	0.044705729	0.044814419813	0.046798243
(1, 1, 1)	0.065740306948	0.063669682	0.064033816183	0.068000835
(3, 2, 5)	0.0068561031151	0.0068327069	0.0068373312816	0.0068797142

5 Certified high-precision values

Because $R_n \rightarrow R_\infty$ only like $1/n^3$, the forward recurrence is useless beyond a few dozen digits. We instead run the *exact downward recurrence*

$$T(m) = T(m+1) + u_m T(m+2), \quad T(\infty) = 1, \quad R_\infty = T(2), \quad (7)$$

where $T(m)$ is the independence polynomial of the tail $\{m, m+1, \dots\}$ (so $T(m) = \sum_{k \geq 0} \sigma_k(m)$ with $\sigma_k(m)$ the order- k cluster sum starting at m). The recurrence (7) is exact, so *all* error lives in the seed $T(M+1), T(M+2)$ at the truncation index M . Truncating the cluster series at order p leaves seed error $\sim \sigma_{p+1} = O(M^{-3(p+1)})$, i.e. about $3(p+1)$ correct digits per decade of M . We use $p = 3$ (~ 12 digits/decade), for which exact closed forms are available:

$$\sigma_2 = \frac{1}{2}(p_1^2 - p_2) - a_1, \quad (8)$$

$$\sigma_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3) - a_1p_1 + c + t, \quad (9)$$

with power sums $p_j = \sum_{n \geq m} u_n^j$ (and $p_1 = \sigma_1(m) = S(m)$ from Proposition 3, exact via ψ), and adjacency sums $a_1 = \sum_n u_n u_{n+1}$, $c = \sum_n u_n u_{n+1}(u_n + u_{n+1})$, $t = \sum_n u_n u_{n+1} u_{n+2}$.

Proposition 4 (Cluster closed forms; σ_2, σ_3 inclusion–exclusion cores PROVEN (Lean, Theorems 1, 2), VERIFIED to $\sim 10^{-54}$). *Equations (8)–(9) hold: the unrestricted elementary symmetric sums $e_2 = \frac{1}{2}(p_1^2 - p_2)$, $e_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3)$ corrected for adjacency by inclusion–exclusion (Appendix A). They agree with the brute-force enumeration of non-consecutive 2- and 3-subsets for (A, B, C) equal to (1, 0, 1), (1, 1, 1), (2, 1, 3), (1, 2, 1) to better than 10^{-40} (absolute residuals $\sim 10^{-54}$ – 10^{-55}).*

This is the gate of `verify_clusters.py`: no derived combinatorial identity is trusted until machine-checked against brute force. With the gate passed, the downward recurrence (7) seeded by $1 + \sigma_1(M) + \sigma_2(M) + \sigma_3(M)$ gives the values in Table 2.

What “certified” means. Throughout, “certified to N digits” is a *concordance* claim, not a proven error bound: the leading N digits are stable under (a) seed-order independence, (b) self-convergence in the truncation M , and (c) an independent forward/Neville extrapolation, and are consistent with the seed-truncation estimate $O(M^{-3(p+1)}) = O(M^{-12})$ for the order- $p=3$ seed.

For the reference triple (1, 0, 1) we certify δ *three* independent ways at $M = 10^6$ (working precision 170 digits): seed-order independence (order-2 vs order-3 seed agree to 56 digits; order-3 vs order-3-plus-leading- σ_4 to 75 digits), self-convergence ($M = 10^5$ vs 10^6 agree to 47 digits), and a fully independent forward continuant with Neville extrapolation in $1/N$ (agrees to ~ 25 digits, its own resolution ceiling). The two cluster-based certificates share the downward-recurrence machinery,

Table 2: Certified high-precision $\delta(A, B, C) = \log R_\infty$ via the exact order-3 cluster seed and downward recurrence (7) at $M = 10^5$ (working precision 90 digits), each value self-checked against $M = 2.5 \times 10^4$ and cross-checked by independent forward/Neville extrapolation. “rel. dig.” is the conservatively certified number of correct digits.

(A, B, C)	$\delta = \log R_\infty$ (first 30 digits)	rel. dig.
(1, 0, 1)	0.123857194360626392728504989702	44 [†]
(1, 0, 2)	0.077741108480647487187139168954	39
(1, 0, 3)	0.055023822866781361284024549668	39
(2, 0, 1)	0.044814419812755566106936562111	40
(3, 0, 1)	0.023024249675539505266856933463	41
(1, 1, 1)	0.064033816182909523022426151229	39
(1, 2, 1)	0.039254537227423694454156302525	39
(2, 1, 3)	0.017942479040119357169479600823	40
(3, 2, 5)	0.006837331281558730523182815470	41

[†] certified to ≥ 44 digits at $M = 10^6$, dps= 170; see (10).

so the ≥ 47 -digit agreement is *internal* concordance among methods sharing that machinery; the only *fully independent* route (forward/Neville) confirms ~ 25 digits, the honest floor on cross-*method* confirmation. The conservative reliable count is ≥ 44 digits:

$$\delta(1, 0, 1) = 0.12385719436062639272850498970259084096757955 \dots \quad (10)$$

Two robustness notes. The degenerate triple (1, 2, 1) has coincident poles, so the simple-pole residue form (4) is singular there; the seed’s $\sigma_1(m) = S(m)$ must use the exact confluent polygamma form (5) (a direct numerical tail sum loses precision at large m). With that substitution (1, 2, 1) matches the others (39 digits), and its $S(2)$ reproduces (6) exactly. Separately, the independent forward/Neville cross-check must start the scaled continuant at the index that includes the u_2 term, or it converges to a different limit; with the correct start it agrees with the cluster value to ~ 25 digits across all tested triples.

On the previously reported “80-digit” figure. An earlier internal record cited an 80-digit value for $\delta(1, 0, 1)$. The present method reaches 80 digits in principle at $M = 10^7$ (consistent with the ~ 12 digit/decade rate), but we *certify* only the ≥ 44 digits of (10) at feasible cost; nothing is asserted beyond the certified count.

6 No low-height closed form for δ

Proposition 5 (Non-elementarity; CONJECTURED, rigorous null). *No low-height integer relation is found for V , R_∞ , or δ against natural constant bases. Specifically: `identify`(V) = None at 100 digits (the strongest null, V being the natural “answer”, known to 129 digits via the fast p_n/q_n ratio); `identify`(δ) = None; PSLQ over $\{1, \pi, \log 2, \gamma, \text{Catalan}, \zeta(3)\}$ with $\max |c_i| \leq 200$ returns no relation; and R_∞ — the natural algebraic candidate, tested by PSLQ on $\{1, R_\infty, R_\infty^2, R_\infty^3, R_\infty^4\}$ — yields no integer relation of degree ≤ 4 and height $\leq 10^4$ (δ and V receive the *identify* null above).*

Uncontrolled PSLQ does return high-height vectors, but `pslq_stability.py` shows them to be artifacts: the recovered vector wanders with working precision ($[30, 110, \dots] \rightarrow [-244, \dots] \rightarrow$

$[1330, \dots] \rightarrow [-25296, \dots]$) and the residual floors at $\sim 10^{-\text{dps}}$ rather than shrinking toward true zero — whereas a genuine integer relation is precision-invariant. We therefore record δ as conjecturally a genuine new transcendental constant of the quadratic family, not reducible to the product constant K_Γ . Irrationality of δ is open. A null integer-relation result is, we stress, a complete and honest answer, not a gap.

7 The machine-checked finitary core

The finitary facts underlying Sections 2–4 are formalized in Lean 4 with Mathlib (toolchain pinned: `lean4:v4.30.0`, `Mathlib rev=v4.30.0`). Each declaration was checked with `#print axioms`; every axiom cone is a subset of the Mathlib defaults `{propext, Classical.choice, Quot.sound}` with `no sorryAx`, the source contains no `sorry/admit`, and the build is warning- and error-free (exit 0).

Theorem 1 (Formalized core; PROVEN). *The following are machine-checked with clean axiom cones:*

- *casoratian_step, casoratian_eq, pq_casoratian*: the Casoratian of two solutions of $s_{n+2} = b_{n+2}s_{n+1} + s_n$ flips sign each step, hence equals $(-1)^n$ times its initial value; for the convergents ($p_0=1, p_1=b_1+1, q_0=1, q_1=b_1$) this is $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$ (over an arbitrary *CommRing*).
- *rseq_ge_one, rseq_mono, rseq_monotone, rmaj_ge_one, rmaj_mono, rseq_le_rmaj*: the scaled sequence $r_{n+2} = r_{n+1} + u_{n+2}r_n$ ($r_0=r_1=1, u \geq 0$) satisfies $r_n \geq 1$, is monotone, and is dominated by the partial products $\prod_{k=2}^n (1 + u_k)$ (over $\mathbb{R}$).
- *rseq_bddAbove_of_rmaj, rseq_tendsto_ciSup, rseq_tendsto_of_rmaj_bddAbove*: monotone convergence — if the partial products are bounded above (equivalently $\sum u_n < \infty$) then $R_n \rightarrow \sup_n R_n$, i.e. R_∞ exists.
- *psum, psum_nonneg, rseq_ge_one_add_psum, rmaj_mul_one_sub_psum_le_one, rinf_bracket, rseq_tendsto_under_psum_bdd*: the closed-form bracket $1 + S \leq R_\infty \leq 1/(1 - S)$ (Proposition 2), conditional on the partial sums $S_n = \sum_{k=2}^n u_k$ being bounded above and $S = \sup_n S_n < 1$; the finitary core is $1 + S_n \leq R_n$ and $(\prod_{k=2}^n (1 + u_k))(1 - S_n) \leq 1$ (over $\mathbb{R}$).
- *NoTwoConsec.erase, indepPoly_zero, indepPoly_one, filter_not_mem_top, sum_filter_mem_top, indepPoly_succ_succ, indepPoly_eq_rseq*: the independence-polynomial identity (Proposition 1), unconditionally. The weighted independence polynomial of the path graph on $\{2, \dots, n\}$, $\text{indepPoly}(u, n) = \sum_T \prod_{i \in T} u_i$ over no-two-consecutive $T \subseteq \{2, \dots, n\}$, equals R_n ; proved by the deletion–contraction split at the top vertex (over $\mathbb{R}$, no positivity needed).
- *sigma2win_eq*: the finite-window inclusion–exclusion core of the σ_2 closed form (Proposition 4, Appendix A), unconditionally: over any window $[m, M]$, the no-two-consecutive pair sum equals $\sigma_2 = e_2 - a_1$ (every ordered pair is consecutive or not). The Newton reduction $e_2 = \frac{1}{2}(p_1^2 - p_2)$ is not yet formalized (see STRUCTURAL below); the σ_3 inclusion–exclusion core is separately PROVEN in Version 1.4 (*sigma3win_eq*, Theorem 2).

This is the *conditional-core* pattern: the finitary algebra and order structure are proved unconditionally, while the single analytic input — that the partial products are bounded above — is taken as an explicit labelled hypothesis (it is the convergence of $\sum u_n = O(\sum k^{-4})$, elementary but analytic). Out of scope for the formalization, and proved by hand/numerics above, are: the value of R_∞ , the closed forms for S , the product constant K_Γ and the Stirling/Gamma asymptotics of (1), and the non-elementarity conjecture.

8 Formalized generalizations and refinements

Versions 1.3 and 1.4 extend the machine-checked core of Section 7 in four independent directions, keeping the same conditional-core discipline: all combinatorial and algebraic content is unconditional, and each analytic input is an explicit labelled hypothesis. Every declaration named below was checked with `#print axioms` to a cone within `{propext, Classical.choice, Quot.sound}` (`IsSol.isSolG` depends on *no* axioms at all), with no `sorryAx`, no `sorry/admit` in source, and build exit 0; the harness `Check.lean` now reports 61 such cones (Version 1.4 adds the σ_3 closed form `sigma3win_eq/partition_1` and a strict-monotonicity cone `rseq_strictMono_succ` for the scaled recurrence of Lemma 1).

In mathematical terms, before naming the formal declarations, the four additions are the following. **(A)** The Casoratian (discrete Wronskian) identity holds not only for the classical continuant but for an *arbitrary* second-order linear recurrence $s_{n+2} = c_{n+2}s_{n+1} + d_n s_n$ over any commutative ring: the Casoratian satisfies $W_{n+1} = -d_n W_n$, so it is the running product $\prod_{k < n} (-d_k)$ times its initial value, and the classical alternating $(-1)^n$ is exactly the unit-coefficient ($d \equiv 1$) specialization. **(B)** The independence-polynomial identity for R_∞ is refined into its degree-graded pieces — the order- k cluster sums σ_k — each obeying its own deletion-contraction recurrence; and, new in Version 1.4, the order-3 closed form $\sigma_3 = e_3 - a_1 p_1 + c + t$ is itself proved, by an inclusion-exclusion (bad-triple) decomposition that reconciles the non-termwise cross term $a_1 p_1$ with the restricted sum. **(C)** The coarse bracket of Proposition 2 is sharpened to an m -term enclosure that sandwiches R_∞ between a partial cluster sum L_m and L_m plus an exact geometric tail, collapsing to the bracket at $m = 1$. **(D)** For the $B = 0$ family the weighted sum $\sum A(2m+1)/(\beta_m \beta_{m+1})$ telescopes in closed form by partial fractions, with a clean conditional limit. Each is stated precisely below, followed by the Lean declarations that witness it.

Theorem 2 (v1.3–v1.4 formal additions; PROVEN). *The following hold with clean axiom cones.*

- (A) **General Casoratian over a commutative ring.** `IsSol.isSolG`, `casoratian_stepG`, `casoratian_eqG`, `pq_casoratianG`: over any `CommRing R`, for arbitrary coefficient sequences $c, d : \mathbb{N} \rightarrow R$, every solution of the general second-order linear recurrence $s_{n+2} = c_{n+2}s_{n+1} + d_n s_n$ has Casoratian $W_n = p_n q_{n-1} - p_{n-1} q_n$ obeying $W_{n+1} = -d_n W_n$, hence $W_n = (\prod_{k < n} (-d_k)) W_0$ and $p_{n+1} q_n - p_n q_{n+1} = (\prod_{k < n} (-d_k)) (p_1 q_0 - p_0 q_1)$. Specialized at $d \equiv 1$ (`casoratian_eqG.recovers_classical`), this is the $(-1)^n$ identity of Lemma 2: the v1.2 Casoratian results are exactly its unit-coefficient case (a faithfulness witness for the generalization).
- (B) **Degree-graded independence, the σ_3 window object, and the σ_3 closed form.** `indepKWin`, `indepKWin_succ_succ`, `indepPolyWin_eq_sum_indepKWin`, `sigma3Win_eq`, `sigma3Win_succ_succ`, `sigma3win_eq`, `partition_1`: the independence polynomial of Proposition 1 is split into its degree- k graded pieces $\text{indepKWin}(u, k, a, b) = \sum_{|T|=k} \prod_{i \in T} u_i$ over no-two-consecutive $T \subseteq [a, b]$, each satisfying the deletion-contraction recurrence; summing over k returns `indepPolyWin`. The order-3 piece is the cluster window object $\text{sigma3Win} = \text{indepKWin}(u, 3, \cdot, \cdot)$, whose recurrence is proved. Version 1.4 promotes the σ_3 closed form (9) itself to PROVEN: `sigma3win_eq` (with helper `partition_1`) proves $\sigma_3 = e_3 - a_1 p_1 + c + t$ over any window, e_3 being the unrestricted elementary-symmetric (all-ordered-triples) sum. The cross term $a_1 p_1$ is not a termwise identity; it is reconciled with σ_3 via the bad-triple decomposition $e_3 = \sigma_3 + \Sigma(\mathcal{B})$ and $a_1 p_1 = \Sigma(\mathcal{B}) + t + c$ (Appendix A), a finite `Finset` computation with no analytic input (clean cone). Only the classical Newton reduction $e_3 = \frac{1}{6}(p_1^3 - 3p_1 p_2 + 2p_3)$ is left unformalized (STRUCTURAL); with it, (9) holds in full.

- (C) **An m -term enclosure refining the bracket.** *tsum_geomTail*, *Rinf_enclosure*, *delta_enclosure*, *enclosure_nonvacuous*, *m1_recovers_S_bracket*, *m1_delta_eq_S_bracket*, *H2_zero_forces_sigma0_le_one*: with $L_m = \sum_{k=0}^m \sigma_k$, the partial-sum/geometric-tail sandwich gives $L_m \leq R_\infty \leq L_m + S^{m+1}/(1 - S)$ and, when $L_m > 0$, $\log L_m \leq \delta \leq \log(L_m + S^{m+1}/(1 - S))$, conditional on the two labelled hypotheses (H1) $R_\infty = \sum_k \sigma_k$ and (H2) $0 \leq \sigma_k \leq S^k$. At $m = 1$ both endpoints reduce exactly to the bracket of Proposition 2, and (H2) at $k = 0$ forces $\sigma_0 \leq 1$ — the two points raised in review, now proved rather than observed.
- (D) **Exact weighted telescoping for the $B = 0$ family.** *partialFraction*, *sum_Ico_telescope*, *weighted_telescope*, *weighted_telescope_value*, *weighted_sum_tendsto*: over any field, with $\beta_m = Am^2 + C$ (the $B = 0$ denominator) nonzero on the summation window, the weight $A(2m + 1) = \beta_{m+1} - \beta_m$ makes $A(2m + 1)/(\beta_m \beta_{m+1})$ telescope to the exact finite value

$$\sum_{m=1}^{N-1} \frac{A(2m+1)}{\beta_m \beta_{m+1}} = \frac{1}{A+C} - \frac{1}{AN^2+C},$$

with no characteristic restriction (the 2 only multiplies, never divides). Over $\mathbb{R}$ the infinite sum equals $1/(A+C)$, PROVEN conditional on the single labelled analytic input $1/(AN^2+C) \rightarrow 0$. The unweighted truncation $\sum 1/(\beta_{m-1}\beta_m)$ has, by contrast, no rational closed form in N (Gosper’s algorithm returns no telescope); its coth/digamma limit needs real special functions and is left out of scope, numerically VERIFIED only.

No numerically VERIFIED statement is silently upgraded by these additions — the σ_3 closed form was promoted to PROVEN only with its own clean axiom cone (*sigma3win_eq*, Version 1.4), and the unweighted telescoping limit remains non-PROVEN — and the epistemic ledger *claims.jsonl* records, per result, the grade and the hypotheses under which each conditional statement may be cited downstream.

9 Epistemic status

- **PROVEN** (Lean, clean cones; Theorem 1): the Casoratian identity, the existence of $R_\infty = \lim R_n$ via monotone–bounded convergence, the closed-form bracket $1 + S \leq R_\infty \leq 1/(1 - S)$ (Proposition 2), the independence-polynomial identity (Proposition 1, unconditional), and the finite-window inclusion–exclusion core $\sigma_2 = e_2 - a_1$ of the σ_2 closed form (Proposition 4, unconditional); the bracket and limit are conditional on the elementary analytic input $\sum u_n < \infty$ (and $S < 1$ for the bracket), the independence-polynomial and σ_2 cores require no analytic input. Version 1.3 (Theorem 2) further proves: the general Casoratian over any commutative ring (with the $(-1)^n$ case recovered as a faithfulness witness), the degree-graded independence refinement and the order-3 cluster window object *sigma3Win*, the m -term enclosure $L_m \leq R_\infty \leq L_m + S^{m+1}/(1 - S)$ (conditional on $R_\infty = \sum_k \sigma_k$ and $0 \leq \sigma_k \leq S^k$), and the exact weighted telescoping identity for the $B = 0$ family (its conditional limit resting only on $1/(AN^2 + C) \rightarrow 0$). Version 1.4 (*sigma3win_eq*, *partition_1*) further proves the σ_3 inclusion–exclusion closed form $\sigma_3 = e_3 - a_1 p_1 + c + t$ over any window, unconditional (no analytic input).
- **STRUCTURAL** (complete elementary hand proof; not yet formalized): the scaled recurrence and tail estimate (Lemma 1; its strict-increase and boundedness sub-facts are machine-checked,

`rseq_strictMono_succ`, but the $O(n^{-3})$ tail rate keeps the lemma `STRUCTURAL`; this is a matter of formalization effort rather than a Mathlib obstruction — the rate is the elementary comparison $\sum_{m>n} u_m = O(\int_n^\infty t^{-4} dt) = O(n^{-3})$, whose `Summable`/tail-sum ingredients Mathlib already supplies), the closed forms for S (Proposition 3), and — the only cluster-form step beyond the Lean σ_2, σ_3 inclusion–exclusion cores above — the classical Newton power-sum reductions $e_2 = \frac{1}{2}(p_1^2 - p_2)$, $e_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3)$ (Proposition 4; the inclusion–exclusion derivation of σ_2, σ_3 is Appendix A, now machine-checked). Mathlib does provide Newton’s identities (`MvPolynomial.NewtonIdentities`); the residual work is not the identity but the bridge from the concrete window `Finset` sums e_k, p_k in the variables u_i to Mathlib’s `esymm/psum` formalism, which we defer to a later revision.

- **VERIFIED** (numerically): the cluster identity exactly (`continuant_verify.py`); σ_2, σ_3 and each step of their inclusion–exclusion derivation to $\sim 10^{-54}$ (`verify_clusters.py`, `verify_inclusion_exclusion.py`); S closed forms to 35–62 digits vs direct summation; the bracket on all tabulated triples; and the δ values of Table 2, certified ≥ 39 digits each and ≥ 44 for $(1, 0, 1)$, with an independent cross-check.
- **CONJECTURED**: δ has no low-height elementary closed form (Proposition 5); δ is a new transcendental constant of the family. Irrationality of δ is open.

10 Code and data availability

Every numerical claim is reproduced by the following scripts (`mpmath`): the harness `harness_rinf.py` (two-solution frame, R_∞ , value V); `continuant_verify.py` (exact independence-polynomial check); `S_closed_form.py`, `S_confluent.py` (closed forms for S); `verify_clusters.py` and `verify_inclusion_exclusion.py` (the σ_2, σ_3 gate and the step-by-step inclusion–exclusion check, Proposition 4, Appendix A); `verify_sigma3_finite_window.py` (exact-rational gate for the order-3 window object) and `verify_teslescope_b0.py` (exact-rational gate for the weighted telescoping identity), both Theorem 2; `hp_delta_v2.py` and `finalize_m2.py` (the high-precision δ values and Table 2, output `delta_values_m2.txt`); `verify_forward_crosscheck.py` (the independent cross-check); `hp_probe.py`, `pslq_stability.py` (the integer-relation nulls, Proposition 5); and the Lean package `lean/pcf_continuant` (`Basic.lean`, `Bracket.lean`, `Indep.lean`, `Sigma.lean`, `Topic3.lean`, `Topic5.lean`, `Check.lean`; Theorems 1 and 2). Version 1.0 of this note is archived at Zenodo (version DOI 10.5281/zenodo.20578401, concept DOI 10.5281/zenodo.20578400) as a supplement to the companion growth-law note [1]; versions 1.1–1.2 added the machine-checked closed-form bracket (`rinf_bracket`), the independence-polynomial identity (`indepPoly_eq_rseq`), and the finite-window σ_2 inclusion–exclusion core (`sigma2win_eq`); versions 1.3–1.4 add the generalizations and refinements of Theorem 2 (general Casoratian, degree-graded independence and the order-3 window object, the σ_3 closed form `sigma3win_eq`, the m -term enclosure, and exact weighted telescoping for $B = 0$). The harness `Check.lean` now reports 61 axiom cones, each a subset of `{propext, Classical.choice, Quot.sound}` with no `sorryAx`. The Lean sources and the verification scripts named above accompany the deposit in its public repository, <https://github.com/papanokechi/pcf-delta>.

A Inclusion–exclusion derivation of the cluster closed forms

This appendix proves Equations (8)–(9) by elementary inclusion–exclusion. Both inclusion–exclusion cores are now machine-checked in Lean (the σ_2 core `sigma2win_eq`, Theorem 1; the σ_3 core

$\sigma_3 = e_3 - a_1 p_1 + c + t$ as `sigma3win.eq`, Theorem 2, Version 1.4), so the hand proof below is the human-readable companion to a `PROVEN` result; the brute-force gate of `verify_clusters.py` and the step-by-step gate of `verify_inclusion_exclusion.py` independently confirm every line to $\sim 10^{-54}$.

Fix the tail $\{m, m+1, \dots\}$ with vertex weights $u_i > 0$ and power sums $p_j = \sum_{i \geq m} u_i^j$. Recall from (3) that $\sigma_k = \sum \prod_{i \in T} u_i$ runs over k -subsets T with no two consecutive indices, and let $e_k = \sum_{|T|=k} \prod_{i \in T} u_i$ be the *unrestricted* elementary symmetric sum. Newton's identities give

$$e_2 = \frac{1}{2}(p_1^2 - p_2), \quad e_3 = \frac{1}{6}(p_1^3 - 3p_1 p_2 + 2p_3). \quad (11)$$

Derivation of (8). Each unrestricted pair $\{i < j\}$ is either non-consecutive (counted by σ_2) or consecutive ($j = i+1$); the consecutive pairs sum to $a_1 = \sum_i u_i u_{i+1}$. Hence $\sigma_2 = e_2 - a_1 = \frac{1}{2}(p_1^2 - p_2) - a_1$. $\square$

Derivation of (9). Classify the unrestricted triples $\{i < j < k\}$ by their number of consecutive pairs. Let $\mathcal{B}$ be those with at least one, so $\sigma_3 = e_3 - \Sigma(\mathcal{B})$, where Σ denotes the weighted sum $\sum \prod_{i \in T} u_i$. A triple has either *exactly one* consecutive pair — an adjacent pair plus an isolated third vertex; call this set $\mathcal{B}_1$ — or *two* — a run $\{i, i+1, i+2\}$; call this $\mathcal{B}_2$ — and $\Sigma(\mathcal{B}_2) = \sum_i u_i u_{i+1} u_{i+2} = t$.

Expand the product of the adjacency sum and the first power sum:

$$a_1 p_1 = \left(\sum_i u_i u_{i+1} \right) \left(\sum_r u_r \right) = \sum_i \sum_r u_i u_{i+1} u_r.$$

Split the inner sum according to whether $r \in \{i, i+1\}$. The terms with $r = i$ or $r = i+1$ sum to $\sum_i u_i u_{i+1} (u_i + u_{i+1}) = c$. The remaining terms, with $r \notin \{i, i+1\}$, each form a triple — the adjacent pair $\{i, i+1\}$ together with a distinct third vertex r — whose total we call W , so that $a_1 p_1 = c + W$. In W each member of $\mathcal{B}_1$ is counted once (its unique adjacent pair fixes $(i, i+1)$ and r is the isolated vertex), while each run $\{a, a+1, a+2\}$ in $\mathcal{B}_2$ is counted twice (as the pair $(a, a+1)$ with $r = a+2$, and as the pair $(a+1, a+2)$ with $r = a$). Thus $W = \Sigma(\mathcal{B}_1) + 2t$, and

$$\Sigma(\mathcal{B}) = \Sigma(\mathcal{B}_1) + t = (W - 2t) + t = W - t = a_1 p_1 - c - t.$$

Subtracting from e_3 in (11),

$$\sigma_3 = e_3 - \Sigma(\mathcal{B}) = \frac{1}{6}(p_1^3 - 3p_1 p_2 + 2p_3) - a_1 p_1 + c + t. \quad \square$$

The script `verify_inclusion_exclusion.py` checks each labelled step independently against brute force over a finite window (e_2, e_3 against the elementary sums; a_1 against the adjacent pairs; $a_1 p_1 = c + W$; $W = \Sigma(\mathcal{B}_1) + 2t$; $\Sigma(\mathcal{B}) = \Sigma(\mathcal{B}_1) + t$; and the two closed forms), for $(A, B, C) \in \{(1, 0, 1), (1, 1, 1), (2, 1, 3), (1, 2, 1)\}$, all to $\sim 10^{-54}$.

References

- [1] Papanokechi, *An exact growth constant for the quadratic polynomial continued fraction (structural proof; numerically verified)*, 2026. Establishes the growth law (1) and leaves $\delta = \log R_\infty$ without a closed form. Zenodo concept DOI 10.5281/zenodo.20564681 (<https://doi.org/10.5281/zenodo.20564681>).
- [2] Papanokechi, *Polynomial Continued Fractions: a Proved Logarithmic Ladder, a $4/\pi$ Casoratian Identity, and 482 Irrational Constants*, 2026. Denominator-growth lemma `lem:Qgrowth` and the standard continuant Q_n . Zenodo concept DOI 10.5281/zenodo.19491767 (<https://doi.org/10.5281/zenodo.19491767>).

- [3] W. Gautschi, *Computational aspects of three-term recurrence relations*, SIAM Review **9**(1) (1967), 24–82. DOI 10.1137/1009002 (<https://doi.org/10.1137/1009002>). Standard modern statement and attribution of Pincherle’s theorem (minimal/recessive solutions of three-term recurrences and continued-fraction convergence).
- [4] O. Perron, *Über Summengleichungen und Poincarésche Differenzengleichungen*, Mathematische Annalen **84** (1921), 1–15. DOI 10.1007/BF01458689 (<https://doi.org/10.1007/BF01458689>). Perron’s theorem on the asymptotics of Poincaré difference equations with growing coefficients.
- [5] G. D. Birkhoff and W. J. Trjitzinsky, *Analytic theory of singular difference equations*, Acta Mathematica **60** (1932), 1–89. DOI 10.1007/BF02398269 (<https://doi.org/10.1007/BF02398269>).
- [6] R. Wong and H. Li, *Asymptotic expansions for second-order linear difference equations*, Journal of Computational and Applied Mathematics **41**(1–2) (1992), 65–94. DOI 10.1016/0377-0427(92)90239-T ([https://doi.org/10.1016/0377-0427\(92\)90239-T](https://doi.org/10.1016/0377-0427(92)90239-T)).
- [7] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/>, §3.6 (*Linear difference equations*; minimal/recessive solutions), §5.15 (*Polygamma functions*), and §5.7 (series); cited at section level.
- [8] F. Johansson et al., *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, version 1.3.0, <https://mpmath.org/>. Provides `nsum`, `digamma`, `polygamma`, `identify`, and `pslq` as used above.

AI-assistance disclosure. Prepared with AI assistance (GitHub Copilot; Anthropic Claude) for drafting, computer-algebra verification, and manuscript preparation. All numerical values were obtained by running the scripts named in the Code and Data Availability section; the Lean axiom cones were obtained by `#print axioms`. The author takes responsibility for the content and for the placement of each claim within the four-grade epistemic partition.